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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



**B.Tech Summer / Winter - 2026
(ESE / Make-up / Supplementary) Examination**

Branch: Artificial Intelligence / CSE.
Subject: LLM Fine Tuning & Optimization
Marks: 100

Sem:-VI
Subject Code: AI6M005 / CS6M005
Date of Exam: 07/05/2026 **Time:** 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand the architecture and working principles of Large Language Models.
2. Apply different fine-tuning methods to adapt pre-trained LLMs for specific tasks.
3. Implement parameter-efficient techniques like Adapters and LoRA to optimize resource use.
4. Utilize optimization methods such as quantization and distillation for efficient deployment.
5. Use modern frameworks and tools to train, fine-tune, and evaluate LLMs effectively.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define foundation models.	L1/ CO1	2
Q.2	State the role of tokenizer in LLMs?.	L1/CO1	2
Q.3	Identify the function of self-attention mechanism.	L2/CO1	2
Q.4	List any two limitations of Transformer models.	L1/CO1	2
Q.5	Define full fine-tuning.	L2/CO2	2
Q.6	Classify types of dataset splits used in training.	L2/CO2	2
Q.7	State the purpose of LoRA matrices.	L2/CO2	2
Q.8	Write the concept of prefix tuning in brief.	L3/ CO2	2
Q.9	Give an example of gradient clipping usage.	L2/CO4	2
Q.10	Define pruning in neural networks.	L1/ CO4	2
Q.11	State the purpose of Flash Attention.	L1/ CO5	2
Q.12	Enlist any two evaluation metrics used in NLP tasks.	L1/CO5	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between supervised and self-supervised learning in LLMs.	L4/CO1	4
Q.14	Illustrate the steps involved in dataset annotation for fine-tuning.	L3/CO2	4
Q.15	Discuss the importance of validation datasets in model training.	L2/CO2	4
Q.16	Examine the working of LoRA in parameter-efficient tuning.	L4/CO3	4
Q.17	Write a short note on Adam optimizer.	L3/CO4	4
Q.18	Sketch and explain data parallelism.	L3/CO5	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate encoder-only architecture with suitable diagram.	L4/CO1	7
Q.20	Explain feature-based fine-tuning approach with examples.	L5/CO2	7
Q.21	Analyse adapter-based fine-tuning and its advantages.	L4/CO3	7
Q.22	Examine gradient-based optimization challenges in deep learning.	L4/CO4	7
Q.23	Describe logging and monitoring tools for LLM experiments.	L5/CO5	7
Q.24	Discuss memory bottlenecks in large-scale model training.	L5/CO5	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Rewrite in detail: a) Bias and fairness issues in LLMs b) Data augmentation techniques c) Training efficiency challenges	L5/CO1, CO2	12
Q.26	Explain different optimization algorithms (Adam, Adafactor) and compare their performance in LLM training.	L6/CO4	12
Q.27	Describe modern frameworks and tools used for LLM training and fine-tuning with their advantages.	L6/CO5	12